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Gao et al.

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(54) **ROBOTIC TOOL RECONFIGURATION SYSTEM**
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(57) **ABSTRACT**

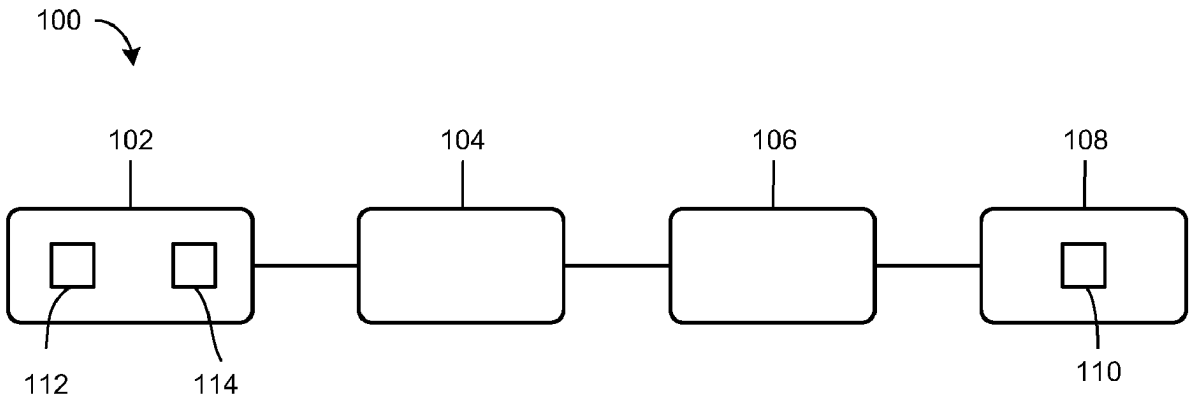
A tool reconfiguration system includes a robot, an end effector, a photogrammetry module, an adjustment module, and a robot control module. The end effector is configured to be attached to the robot and to a vehicle component. The end effector is adjustable between a plurality of configurations. The photogrammetry module includes a first camera that is configured to capture images of the end effector. The adjustment module is configured to receive the images of the end effector and determine whether the end effector has been adjusted from a first configuration to a second configuration based on at least one of the images. The robot control module is configured to, based on the determination, selectively actuate one or more components of the end effector and adjust a present configuration of the end effector to one of the first configuration and the second configuration.

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B25J 9/16 (2006.01)
(52) **U.S. Cl.**
CPC **B25J 9/1697** (2013.01); **B25J 9/1615** (2013.01); **B25J 15/0052** (2013.01)
(58) **Field of Classification Search**
CPC B25J 9/1697; B25J 9/1615; B25J 15/0052; B25J 9/1692; B25J 9/1612; B25J 15/0061; B25J 9/08; B25J 9/161; G01C 11/06; G05B 2219/39468
See application file for complete search history.

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20 Claims, 3 Drawing Sheets



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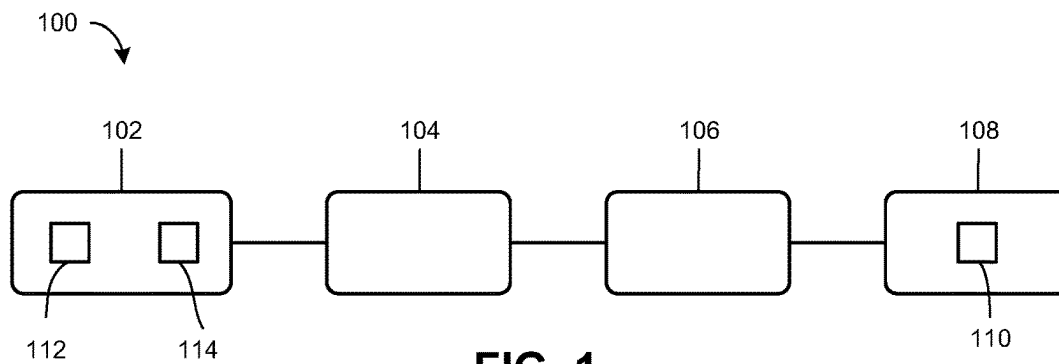


FIG. 1

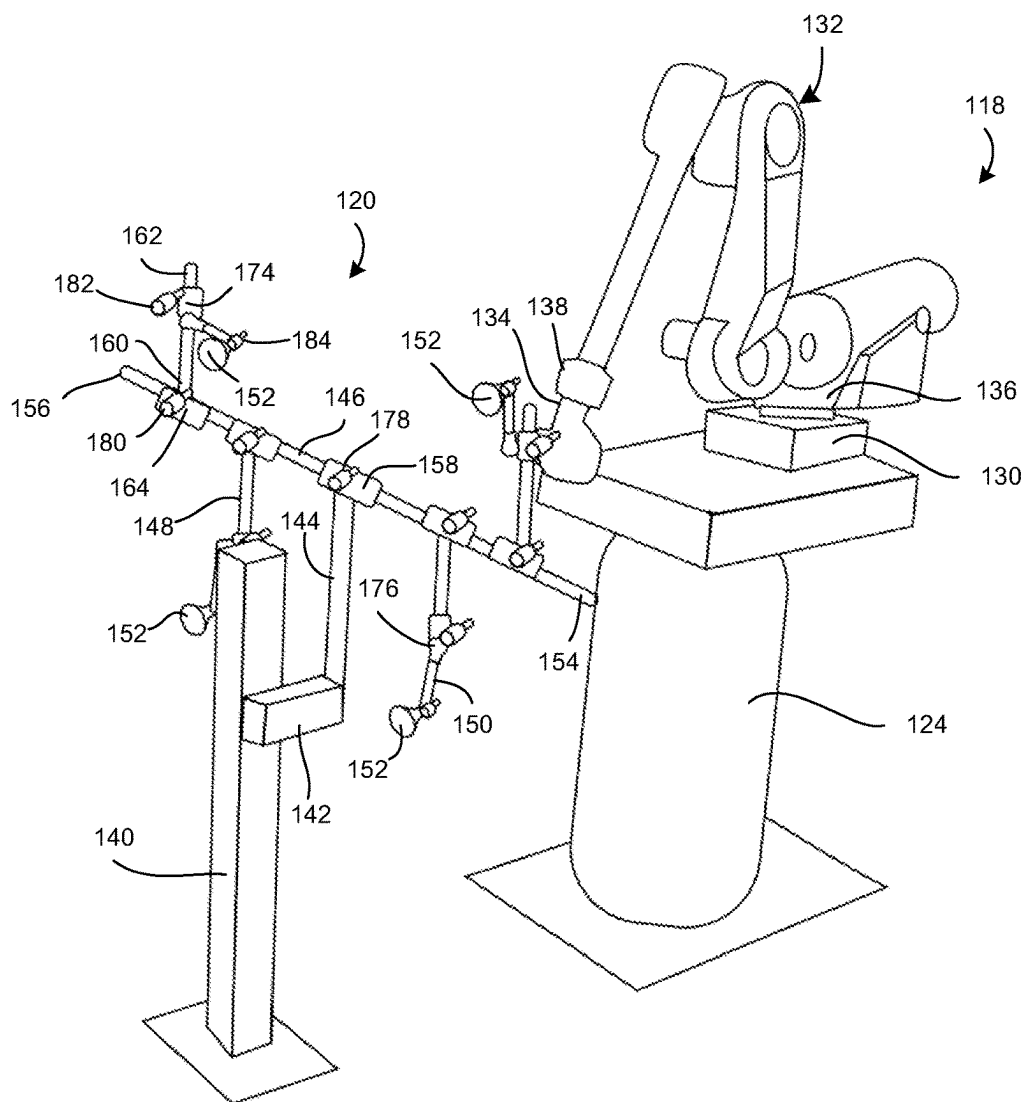


FIG. 2

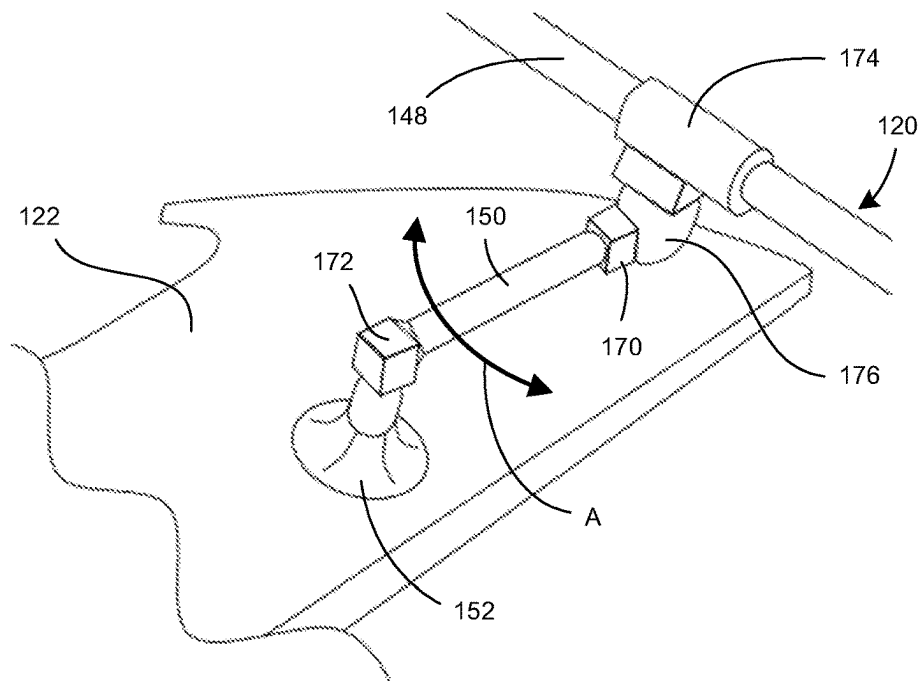


FIG. 3

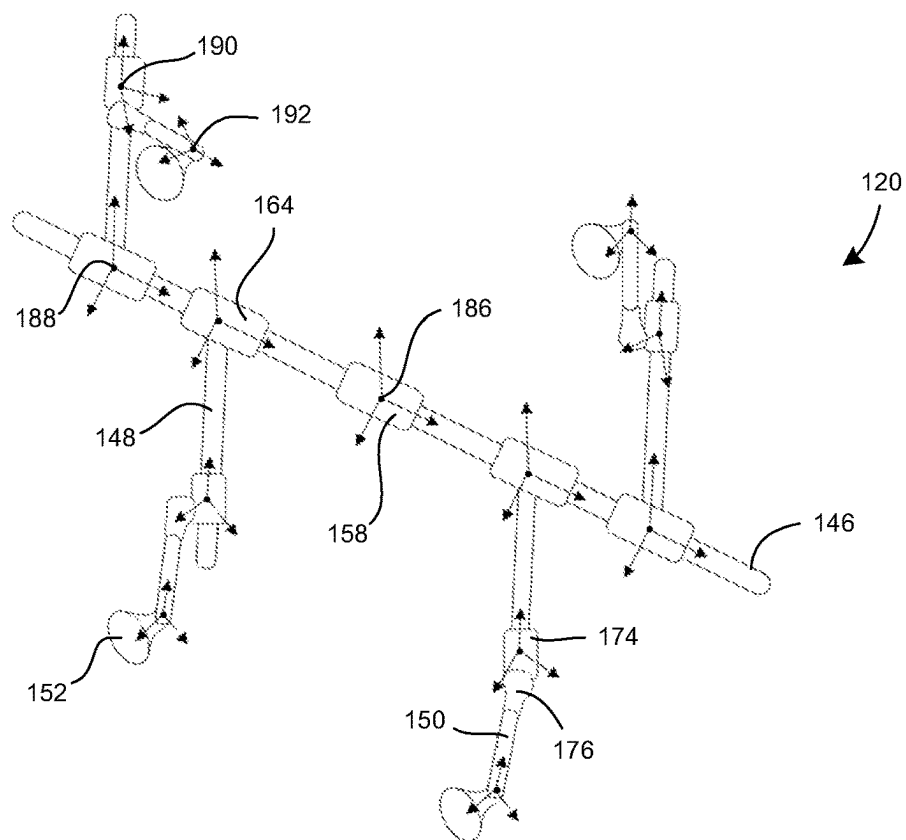


FIG. 4

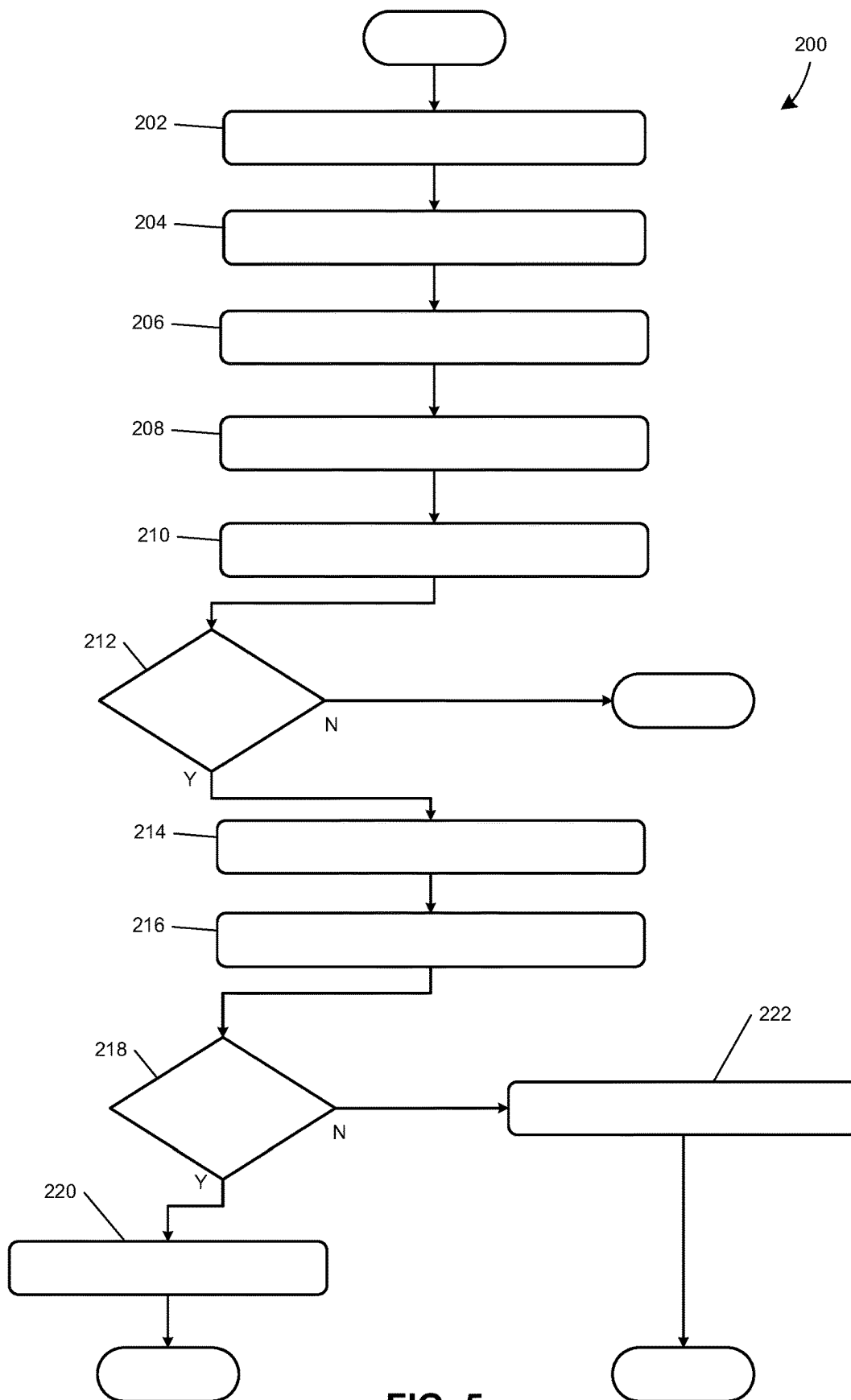


FIG. 5

1

ROBOTIC TOOL RECONFIGURATION SYSTEM

INTRODUCTION

The information provided in this section is for the purpose of generally presenting the context of the disclosure. Work of the presently named inventors, to the extent it is described in this section, as well as aspects of the description that may not otherwise qualify as prior art at the time of filing, are neither expressly nor impliedly admitted as prior art against the present disclosure.

The present disclosure relates to vehicle manufacturing and more particularly to a robotic tool reconfiguration system.

In a vehicle manufacturing plant, robots may transport a vehicle component from one location to another location. The robot may include a robot arm that is movable and an end effector attached to the robot arm. The end effector may attach to the vehicle component and be configurable into various styles based on the specific vehicle component.

SUMMARY

An example of a tool reconfiguration system includes a robot, an end effector, a photogrammetry module, an adjustment module, and a robot control module. The end effector is configured to be attached to the robot and to a vehicle component. The end effector is adjustable between a plurality of configurations. The photogrammetry module includes a first camera that is configured to capture images of the end effector. The adjustment module is configured to receive the images of the end effector and determine whether the end effector has been adjusted from a first configuration to a second configuration based on at least one of the images. The robot control module is configured to, based on the determination, selectively actuate one or more components of the end effector, and adjust a present configuration of the end effector to one of the first configuration and the second configuration.

In one example, the end effector includes two or more identifiers disposed on an outer surface of the end effector and the adjustment module is configured to determine whether the end effector has been adjusted from the first configuration to the second configuration using the identifiers captured in the at least one of the images.

In one example, the adjustment module is configured to determine coordinate systems on the end effector based on positions of the two or more identifiers, respectively. The adjustment module is configured to determine whether the end effector has been adjusted from the first configuration to the second configuration based on a change in the coordinate systems.

In one example, each of the coordinate systems includes an origin point, an X axis, a Y axis, and a Z axis.

In one example, the change includes a change in at least one of an X direction, a Y direction, and a Z direction.

In one example, the change includes a change in angle of at least one of an X direction, a Y direction, and a Z direction.

In one example, the robot control module is configured to store the second configuration in memory in response to receipt of user input indicating the end effector was intentionally adjusted from the first configuration to the second configuration.

In one example, a reconfiguration tool that is configured to actuate one or more of the components of the end effector.

2

In one example, the robot control module is configured to actuate the reconfiguration tool thereby actuating the one or more of the components of the end effector and adjusting the present configuration of the end effector to one of the first configuration and the second configuration.

In one example, the end effector includes a main rod, rod branches adjustably attached to the main rod, and an attachment device adjustably attached to each rod branch.

In one example, the end effector includes a first identifier disposed on the main rod. The adjustment module is configured to determine a first coordinate system on the main rod based on a position of the first identifier.

In one example, at least one rod branch of the rod branches is slidably and rotatably attached to the main rod using a first sleeve. The end effector includes a second identifier disposed on the first sleeve. The adjustment module is configured to determine a second coordinate system on the first sleeve based on a position of the second identifier.

In one example, the adjustment module is configured to determine a first change in the present configuration relative to the first configuration of the first sleeve using the first and second coordinate systems.

In one example, the attachment device is slidably and rotatably attached to the respective rod branch using a second sleeve. The end effector includes a third identifier disposed on the second sleeve. The adjustment module is configured to determine a third coordinate system on the second sleeve based on a position of the third identifier.

In one example, the adjustment module is configured to determine whether the end effector has been adjusted from the first configuration to the second configuration based on the first, second, and third coordinate systems.

In one example, the attachment device is a suction cup.

In one example, the photogrammetry module includes a second camera configured to capture second images. The adjustment module is configured to receive the second images of the end effector and determine whether the end effector has been adjusted from the first configuration to the second configuration based on the images and the second images.

An example of a tool reconfiguration method is provided. The method includes capturing images of an end effector that is adjustable between a plurality of configurations. The method includes determining whether the end effector has been adjusted from a first configuration to a second configuration based on at least one of the images. The method includes selectively actuating one or more components of the end effector and adjusting a present configuration of the end effector to one of the first configuration and the second configuration based on the determination.

In one example, the determining whether the end effector has been adjusted from the first configuration to the second configuration includes determining a first coordinate system based on a position of a first identifier that is disposed on the end effector and captured in the at least one of the images, determining a second coordinate system based on a position of a second identifier that is disposed on the end effector and captured in the at least one of the images, and determining whether the end effector has been adjusted from the first configuration to the second configuration based on a change in the first and second coordinate systems.

In one example, the method includes storing the second configuration in memory in response to receipt of user input indicating the end effector was intentionally adjusted from the first configuration to the second configuration.

Further areas of applicability of the present disclosure will become apparent from the detailed description, the claims

and the drawings. The detailed description and specific examples are intended for purposes of illustration only and are not intended to limit the scope of the disclosure.

BRIEF DESCRIPTION OF THE DRAWINGS

The present disclosure will become more fully understood from the detailed description and the accompanying drawings, wherein:

FIG. 1 is a functional block diagram of an example tool reconfiguration system;

FIG. 2 is perspective view of an example robot and an end effector;

FIG. 3 is a partial perspective view of the example end effector attached to a vehicle component;

FIG. 4 is a perspective view of the example end effector; and

FIG. 5 is a flowchart depicting an example method of the tool reconfiguration system.

In the drawings, reference numbers may be reused to identify similar and/or identical elements.

DETAILED DESCRIPTION

The present application involves a tool reconfiguration system for robots. Robots may include a robot arm and an end effector attached to the robot arm. In a vehicle manufacturing plant, the end effector may pick up and move a vehicle component from one location to another location. The end effector may be adjustable to a variety of different configurations, such as for different vehicle components. For example, attachment devices, such as suction cups, of the end effector may be individually adjustable between different configurations.

In one example, the configuration of the end effector may be adjusted using a hand tool. The configuration of the end effector may be adjusted, for example, such that one or more attachment devices contact a specific vehicle component in one or more improved locations. One or more attachment devices may also naturally move as a result of production or one or more environmental conditions. Examples of environmental conditions include a change in temperature within the vehicle manufacturing plant that causes bending or moving of one or more components of the end effector.

The attachment devices of the end effector may change without knowledge by the robot or a control module of the robot. Without the knowledge of movement, however, one or more collisions may occur. A tool reconfiguration system is needed in light of these challenges to identify whether the configuration of the end effector has changed and if so, to notify the robot control module of the present configuration.

FIG. 1 is a functional block diagram of a tool reconfiguration system 100. The tool reconfiguration system 100 may be disposed within a vehicle manufacturing plant of a vehicle original equipment manufacturer (OEM). The tool reconfiguration system 100 may include a photogrammetry module 102 (e.g., including a photogrammetry assembly), an adjustment module 104, a robot control module 106, and a robot 108 that is attachable to an end effector 110.

The photogrammetry module 102 may include one or more vision sensors (e.g., cameras) configured to capture vision data (e.g., images) of the robot 108. The vision sensors are positioned within the vehicle manufacturing plant to visually capture identifiers that are disposed on the end effector 110. Examples of identifiers include fiducials (e.g., reflective), quick response (QR) codes, markers, or another suitable type of visual identifier. The photogram-

metry module 102 transmits the vision data to the adjustment module 104. The vision data may be transmitted by wire or wirelessly.

In the illustrated example, the photogrammetry module 102 includes a first camera 112 and a second camera 114 configured to capture first images and second images, respectively. The first camera 112 and the second camera 114 may be disposed on one side of the end effector 110 such that the first camera 112 is spaced from the second camera 114. Alternatively, the first camera 112 may be disposed on a first side of the end effector 110 and the second camera 114 may be disposed on a second side of the end effector 110. In one example, the second side may be opposite the first side, such as left to right. In another example, the second side may be perpendicular to, or any other suitable angle away from, the first side. The first and second cameras 112, 114 are configured to capture a field of view ranging from about 10 cubic meters (m³) to about 15 m³. The first and second cameras 112, 114 capture images at a rate of about 100 hertz (Hz) or another suitable frequency and a level of accuracy of about 0.1 millimeter (mm) or another suitable accuracy.

The adjustment module 104 defines a series of coordinate systems on the end effector 110 of the robot 108. Each coordinate system includes an origin point, an X-axis, a Y-axis, and a Z axis. In one example, the series of coordinates systems includes a reference coordinate system that may be defined based on a fixed component of the end effector 110 and additional coordinate systems that may be defined based on locations of adjustable components of the end effector 110. The adjustment module 104 is configured to determine whether the end effector 110 has been adjusted from an original configuration to a present configuration using the series of coordinate systems. If the end effector 110 had been adjusted, the adjustment module 104 automatically calculates the amount and direction of the adjustment (X direction, Y direction, Z direction, angular direction, etc.) and automatically generates a vision-servoing command based on the same. The vision-servoing command may be transmitted to the robot control module 106 by wire or wirelessly.

The robot control module 106 converts the vision-servoing command into a robot command and actuates the robot 108 based on the robot command. The robot command may be automatically transmitted to the robot 108 by wire or wirelessly. In this manner, the robot command is adjusted based on one or more differences between the original configuration and the present configuration.

With additional reference to FIGS. 2-4, an example robot 118 with an example end effector 120 is shown. FIG. 2 is a perspective view of the example robot 118. FIG. 3 is a partial perspective view of the example end effector 120 attached to a vehicle component 122. FIG. 4 is perspective view of the example end effector 120.

The example robot 118 is a 6 degrees of freedom (DOF) robot. However, any suitable robot may be used, such as a 7 DOF or higher robot. Additionally, the example end effector 120 is a 5 DOF end effector. However, the end effector 120 may be any suitable end effector, such as a 6 DOF end effector.

The robot 118 may be disposed on a first stand 124, as illustrated, or alternatively, disposed on a floor surface. The robot 108 may include a robot base 130, a robot arm 132, and a reconfiguration tool 134. The end effector 120 is connectable to and disconnectable from the robot 108. The robot arm 132 extends between a first end 136 and a second end 138 that is opposite the first end 136. The robot arm 132 may be attached to the robot base 130 at the first end 136 and

5

coupled to the end effector **120** at the second end **138**. The robot arm **132** is movable relative to the robot base **130**.

The second end **138** of the robot arm **132** may be attached to the reconfiguration tool **134** and configured to interact with the end effector **120**. The reconfiguration tool **134** may be configured to change the configuration of the end effector **120** by adjusting one or more components (e.g., branches, arms, attachment devices, etc.) of the end effector **120**. In the illustrated example of FIG. 2, the reconfiguration tool **134** is a pneumatic torque wrench, referred to as a nutrunner. However, one or more other suitable reconfiguration tools may additionally or alternatively be used. The reconfiguration tool **134** may be connectable to and disconnectable from the robot arm **132**. For example, the reconfiguration tool **134** may be disconnected from the robot arm **132** after adjusting the end effector **120** to one configuration.

The second end **138** of the robot arm **132** may be attached and connected to the end effector **120** and configured to move the end effector **120**. For example, FIG. 2 illustrates the end effector **120** sitting on a second stand **140** using a tool changer **142**. The configuration of the end effector **120** may be adjusted using the reconfiguration tool **134** while the end effector **120** rests on the second stand **140**.

FIG. 3 illustrates the end effector **120** holding the vehicle component **122**. The robot arm **132** may move the end effector **120** from the second stand **140** to the vehicle component **122** and actuate one or more components of the end effector **120** to hold the vehicle component **122**. The vehicle component **122** is illustrated as a vehicle hood. However, the vehicle component **122** may be other vehicle components. The present application is also applicable to non-vehicle components.

The end effector **120** includes a support rod **144**, a main rod **146**, rod branches **148** that extend from the main rod **146**, a rod arm **150** that extends from each rod branch **148**, and an attachment device **152** disposed at each rod arm **150**. In the illustrated example, the rod branches **148** includes four rod branches. However, a greater or lesser number of rod branches **148** may be used. Additionally, the attachment device **152** is a suction cup in the illustrated example. However, one or more other suitable types of attachment devices may be used.

The main rod **146** extends linearly between a third end **154** and a fourth end **156** that is opposite the third end **154**. A connector **158** may be disposed between the third and fourth ends **154**, **156**. The connector **158** may be used to secure the end effector **120** to the support rod **144**. The second end **138** of the robot arm **132** may connect to the connector **158** to move and actuate the end effector **120**. The attachment devices **152** may be actuated to hold the vehicle component **122** pneumatically or in another suitable manner. For example, the robot **118** may draw air from within the suction cups to hold the vehicle component **122** using the attachment devices **152**.

Each rod branch **148** may extend between a fifth end **160** and a sixth end **162** that is opposite the fifth end **160**. The fifth end **160** is attached to the main rod **146** using a first sleeve **164**. The first sleeve **164** may extend annularly about the main rod **146**. The first sleeve **164** is configured to be slidable (in this example horizontally) along the main rod **146** and rotatable about the main rod **146**. Each rod branch **148** may be substantially perpendicular to the main rod **146**.

Each rod arm **150** may extend between a seventh end **170** and an eighth end **172** that is opposite the seventh end **170** (FIG. 3). The seventh end **170** is attached to the rod branch **148** using a second sleeve **174** and a swing connector **176**. The second sleeve **174** may extend annularly about the rod

6

branch **148**. The second sleeve **174** is configured to be slidable (in this example vertically) along the rod branch **148** and rotatable about the rod branch **148**. The swing connector **176** is coupled to the second sleeve **174** and attached to the seventh end **170**. The swing connector **176** is configured to swing (i.e., pivot) the rod arm **150** relative to the rod branch **148** in an angular direction **A**. The eighth end **172** is attached to the attachment device **152**.

Identifiers are disposed on an outer surface of the end effector **120**. In the illustrated example shown in FIG. 2, a first identifier **178** is disposed on the main rod **146**, a second identifier **180** is disposed on the first sleeve **164**, a third identifier **182** is disposed on a second sleeve **174**, and a fourth identifier **184** is disposed on the eighth end **172** of the rod arm **150**. However, a greater or lesser number of identifiers may be used, and the identifiers may be disposed on other locations on the end effector **120**. In some configurations, identifiers may also be disposed on static components within the vehicle manufacturing plant, such as on a floor or wall surface.

As shown in FIG. 4, a series of coordinate systems may be determined visually on the end effector **120** from the first and second images. The system of coordinates may include a first coordinate system **186**, a second coordinate system **188**, a third coordinate system **190** and a fourth coordinate system **192**. The first coordinate system **186** is determined in a location of the first identifier **178** on the main rod **146**. The first coordinate system **186** may be positioned at the connector **158** or an alternative position on the main rod **146** that is non-adjustable. The second coordinate system **188** is determined in a location of the second identifier **180** on the first sleeve **164**. The third coordinate system **190** is determined in a location of the third identifier **182** on the second sleeve **174**. The fourth coordinate system **192** is defined in a location of the fourth identifier **184** on the eighth end **172** of the rod arm **150**. The first coordinate system **186** serves as a reference coordinate system for the end effector **120** and the second, third, and fourth coordinate systems **188**, **190**, **192** may be determined for each of the attachment devices **152** of the end effector **120**.

FIG. 5 is a flowchart depicting an example method **200** of the tool reconfiguration system **100** using the example end effector **120**.

At **202**, the first and second cameras **112**, **114** of the photogrammetry module **102** capture first and second images of the identifiers **178**, **180**, **182**, **184** of the end effector **120** and transmit the first and second images to the adjustment module **104**.

At **204**, the adjustment module **104** detects the identifiers **178**, **180**, **182**, **184** within the first and second images and determines a location of the first coordinate system **186**, the second coordinate system **188**, the third coordinate system **190**, and the fourth coordinate system **192** on the end effector **120** based on the locations and orientations of the respective identifiers **178**, **180**, **182**, **184**. The adjustment module **104** determines the present configuration of the end effector **120** based on the series of coordinate systems.

At **206**, the adjustment module **104** determines a first change in the present configuration of the end effector **120** relative to the original configuration of the end effector **120** using the first and second coordinate systems **186**, **188**. The original configuration of the end effector **120** may be stored in memory. More specifically, the adjustment module **104** determines whether the first sleeve **164** has slidably moved along the main rod **146** relative to an original position of the first sleeve **164** on the main rod **146** in the original configuration. If the first sleeve **164** has moved, the adjustment

module **104** calculates an amount of movement and a direction of movement along the main rod **146** based on the position of the first and second coordinate systems **186**, **188** in the present configuration relative to the position of the first and second coordinate systems **186**, **188** in the original configuration.

Additionally, the adjustment module **104** determines whether the first sleeve **164** has rotated about the main rod **146** relative to a rotation of the first sleeve **164** in the original configuration. If the first sleeve **164** has rotated, the adjustment module **104** calculates an amount of rotation and a direction of rotation about the main rod **146** based on the position of the first and second coordinate systems **186**, **188** in the present configuration relative to the position of the first and second coordinate systems **186**, **188** in the original configuration. In other words, the first change in the present configuration of the end effector **120** relative to the original configuration includes the amount of movement and direction of movement of the first sleeve **164** along the main rod **146** and the amount of rotation and direction of rotation about the main rod **146**.

At **208**, the adjustment module **104** determines a second change in the present configuration of the end effector **120** relative to the original configuration of the end effector **120** using the second and third coordinate systems **188**, **190**. More specifically, the adjustment module **104** determines whether the second sleeve **174** has slidably moved along the rod branch **148** relative to an original position of the second sleeve **174** on the rod branch **148** in the original configuration. If the second sleeve **174** has moved, the adjustment module **104** calculates an amount of movement and a direction of movement along the rod branch **148** based on the position of the second and third coordinate systems **188**, **190** in the present configuration relative to the position of the second and third coordinate systems **188**, **190** in the original configuration.

Additionally, the adjustment module **104** determines whether the second sleeve **174** has rotated about the rod branch **148** relative to a rotation of the second sleeve **174** in the original configuration. The adjustment module **104** calculates an amount of rotation and a direction of the rotation about the rod branch **148** based on the position of the second and third coordinate systems **188**, **190** in the present configuration relative to the position of the second and third coordinate systems **188**, **190** in the original configuration. In other words, the second change in the present configuration of the end effector **120** relative to the original configuration of the end effector **120** includes the amount of movement and direction of movement of the second sleeve **174** along the rod branch **148** and the amount of rotation and direction of rotation about the rod branch **148**.

At **210**, the adjustment module **104** determines a third change in the present configuration of the end effector **120** relative to the original configuration of the end effector **120** using the third and fourth coordinate systems **190**, **192**. More specifically, the adjustment module **104** determines whether the eighth end **172** of the rod arm **150** has rotated relative to an original position of the second arm end **174** in the original configuration. If the eighth end **172** has rotated, the adjustment module **104** calculates an amount of rotation and a direction of the rotation based on the position of the third and fourth coordinate systems **190**, **192** in the present configuration relative to the position of the third and fourth coordinate systems **190**, **192** in the original configuration. In other words, the third change in the present configuration of the end effector **120** relative to the original configuration of

the end effector **120** includes the amount and direction of rotation of the eighth end **172** of the rod arm **150**.

At **212**, the adjustment module **104** determines whether the first, second, and third changes in the present configuration of the end effector **120** relative to the original configuration of the end effector **120** exceed a predetermined tolerance. If **212** is false, the method **200** ends. If **212** is true, the method **200** continues to **214**.

At **214**, the adjustment module **104** automatically generates a vision-servoing command using the first, second, and third changes in the present configuration of the end effector **120** relative to the original configuration of the end effector **120**. The adjustment module **104** transmits the vision-servoing command to the robot control module **106**.

At **216**, the robot control module **106** generates a robot command using the vision-servoing command. More specifically, the robot control module **106** processes the first, second, and third changes and determines a global change (i.e., absolute change) in the present configuration of the attachment device **152** relative to the original configuration of the attachment device **152**. The global change includes an amount and direction of change of the attachment device **152** relative to the first coordinate system **186**.

At **218**, the robot control module **106** determines whether the global change of the attachment device **152** from the original configuration to the present configuration was intentional. In some examples, the robot control module **106** may request user input indicative of whether the change was intentional.

If **218** is true (e.g., when user input is received indicative of the change being intentional), the method continues to **220**. At **220**, the robot control module **106** stores the present configuration for the given vehicle component **122** and transmits the robot command to the robot **118**. Using the robot command, the robot arm **132** positions the reconfiguration tool **134** to the correct location of the attachment devices **152** of the end effector **120**, for example, to change the configuration of the end effector **120** for a different vehicle component. Additionally, for future interactions with the given vehicle component **122**, the robot control module **106** actuates the end effector **120** to the present configuration.

If **218** is false, the method **200** continues to **222**. At **222**, the robot control module **106** transmits the robot command to the robot **118**. Using the robot command, the robot **118** positions the reconfiguration tool **134** to the present location of the attachment devices **152** and adjusts the attachment devices **152**. In one example, the reconfiguration tool **134** adjusts the attachment devices **152** to revert the end effector **120** from the present configuration back to the original configuration. For future interactions with the vehicle component **122**, the robot control module **106** actuates the end effector **120** to the original configuration. In another example, the reconfiguration tool **134** adjust the attachment devices **152** of the end effector **120** to change the configuration of the end effector **120** for a different vehicle component.

The foregoing description is merely illustrative in nature and is in no way intended to limit the disclosure, its application, or uses. The broad teachings of the disclosure can be implemented in a variety of forms. Therefore, while this disclosure includes particular examples, the true scope of the disclosure should not be so limited since other modifications will become apparent upon a study of the drawings, the specification, and the following claims. It should be understood that one or more steps within a method may be executed in different order (or concurrently) without

altering the principles of the present disclosure. Further, although each of the embodiments is described above as having certain features, any one or more of those features described with respect to any embodiment of the disclosure can be implemented in and/or combined with features of any of the other embodiments, even if that combination is not explicitly described. In other words, the described embodiments are not mutually exclusive, and permutations of one or more embodiments with one another remain within the scope of this disclosure.

Spatial and functional relationships between elements (for example, between modules, circuit elements, semiconductor layers, etc.) are described using various terms, including “connected,” “engaged,” “coupled,” “adjacent,” “next to,” “on top of,” “above,” “below,” and “disposed.” Unless explicitly described as being “direct,” when a relationship between first and second elements is described in the above disclosure, that relationship can be a direct relationship where no other intervening elements are present between the first and second elements, but can also be an indirect relationship where one or more intervening elements are present (either spatially or functionally) between the first and second elements. As used herein, the phrase at least one of A, B, and C should be construed to mean a logical (A OR B OR C), using a non-exclusive logical OR, and should not be construed to mean “at least one of A, at least one of B, and at least one of C.”

In the figures, the direction of an arrow, as indicated by the arrowhead, generally demonstrates the flow of information (such as data or instructions) that is of interest to the illustration. For example, when element A and element B exchange a variety of information but information transmitted from element A to element B is relevant to the illustration, the arrow may point from element A to element B. This unidirectional arrow does not imply that no other information is transmitted from element B to element A. Further, for information sent from element A to element B, element B may send requests for, or receipt acknowledgements of, the information to element A.

In this application, including the definitions below, the term “module” or the term “controller” may be replaced with the term “circuit.” The term “module” may refer to, be part of, or include: an Application Specific Integrated Circuit (ASIC); a digital, analog, or mixed analog/digital discrete circuit; a digital, analog, or mixed analog/digital integrated circuit; a combinational logic circuit; a field programmable gate array (FPGA); a processor circuit (shared, dedicated, or group) that executes code; a memory circuit (shared, dedicated, or group) that stores code executed by the processor circuit; other suitable hardware components that provide the described functionality; or a combination of some or all of the above, such as in a system-on-chip.

The module may include one or more interface circuits. In some examples, the interface circuits may include wired or wireless interfaces that are connected to a local area network (LAN), the Internet, a wide area network (WAN), or combinations thereof. The functionality of any given module of the present disclosure may be distributed among multiple modules that are connected via interface circuits. For example, multiple modules may allow load balancing. In a further example, a server (also known as remote, or cloud) module may accomplish some functionality on behalf of a client module.

The term memory circuit is a subset of the term computer-readable medium. The term computer-readable medium, as used herein, does not encompass transitory electrical or electromagnetic signals propagating through a medium

(such as on a carrier wave); the term computer-readable medium may therefore be considered tangible and non-transitory. Non-limiting examples of a non-transitory, tangible computer-readable medium are nonvolatile memory circuits (such as a flash memory circuit, an erasable programmable read-only memory circuit, or a mask read-only memory circuit), volatile memory circuits (such as a static random access memory circuit or a dynamic random access memory circuit), magnetic storage media (such as an analog or digital magnetic tape or a hard disk drive), and optical storage media (such as a CD, a DVD, or a Blu-ray Disc).

The apparatuses and methods described in this application may be partially or fully implemented by a special purpose computer created by configuring a general purpose computer to execute one or more particular functions embodied in computer programs. The functional blocks, flowchart components, and other elements described above serve as software specifications, which can be translated into the computer programs by the routine work of a skilled technician or programmer.

The computer programs include processor-executable instructions that are stored on at least one non-transitory, tangible computer-readable medium. The computer programs may also include or rely on stored data. The computer programs may encompass a basic input/output system (BIOS) that interacts with hardware of the special purpose computer, device drivers that interact with particular devices of the special purpose computer, one or more operating systems, user applications, background services, background applications, etc.

The computer programs may include: (i) descriptive text to be parsed, such as HTML (hypertext markup language), XML (extensible markup language), or JSON (JavaScript Object Notation) (ii) assembly code, (iii) object code generated from source code by a compiler, (iv) source code for execution by an interpreter, (v) source code for compilation and execution by a just-in-time compiler, etc. As examples only, source code may be written using syntax from languages including C, C++, C #, Objective-C, Swift, Haskell, Go, SQL, R, Lisp, Java®, Fortran, Perl, Pascal, Curl, OCaml, Javascript®, HTML5 (Hypertext Markup Language 5th revision), Ada, ASP (Active Server Pages), PHP (PHP: Hypertext Preprocessor), Scala, Eiffel, Smalltalk, Erlang, Ruby, Flash®, Visual Basic®, Lua, MATLAB, SIMULINK, and Python®.

What is claimed is:

1. A tool reconfiguration system comprising:

- a robot;
- an end effector configured to be attached to the robot and to a vehicle component, the end effector being adjustable between a plurality of configurations of suction cups;
- a photogrammetry module including a first camera that is configured to capture images of the end effector;
- an adjustment module configured to:
 - receive the images of the end effector; and
 - determine whether the end effector has been adjusted from a first configuration of the suction cups to a second configuration of the suction cups based on at least one of the images; and
- a robot control module configured to, in response to a determination that the end effector has been adjusted from the first configuration of the suction cups to the second configuration of the suction cup, selectively actuate one or more components of the end effector and

11

adjust a present configuration of the suction cups of the end effector to one of the first configuration of the suction cups.

2. The tool reconfiguration system of claim 1, wherein: the end effector includes two or more identifiers disposed on an outer surface of the end effector; and the adjustment module is configured to determine whether the end effector has been adjusted from the first configuration to the second configuration using the identifiers captured in the at least one of the images.
3. The tool reconfiguration system of claim 2, wherein: the adjustment module is configured to determine coordinate systems on the end effector based on positions of the two or more identifiers, respectively; and the adjustment module is configured to determine whether the end effector has been adjusted from the first configuration to the second configuration based on a change in the coordinate systems.
4. The tool reconfiguration system of claim 3, wherein each of the coordinate systems includes an origin point, an X axis, a Y axis, and a Z axis.
5. The tool reconfiguration system of claim 3, wherein the change includes a change in at least one of an X direction, a Y direction, and a Z direction.
6. The tool reconfiguration system of claim 3, wherein the change includes a change in angle of at least one of an X direction, a Y direction, and a Z direction.
7. The tool reconfiguration system of claim 1, the robot control module is configured to store the second configuration in memory in response to receipt of user input indicating the end effector was intentionally adjusted from the first configuration to the second configuration.
8. The tool reconfiguration system of claim 1, further comprising a reconfiguration tool that is configured to actuate one or more of the components of the end effector.
9. The tool reconfiguration system of claim 8, wherein the robot control module is configured to actuate the reconfiguration tool thereby actuating the one or more of the components of the end effector and adjusting the present configuration of the end effector to the first configuration.
10. The tool reconfiguration system of claim 1, wherein the end effector includes a main rod, rod branches adjustably attached to the main rod, and an attachment device adjustably attached to each rod branch.
11. The tool reconfiguration system of claim 10, wherein: the end effector includes a first identifier disposed on the main rod; and the adjustment module is configured to determine a first coordinate system on the main rod based on a position of the first identifier.
12. The tool reconfiguration system of claim 11, wherein: at least one rod branch of the rod branches is slidably and rotatably attached to the main rod using a first sleeve; the end effector includes a second identifier disposed on the first sleeve; and the adjustment module is configured to determine a second coordinate system on the first sleeve based on a position of the second identifier.
13. The tool reconfiguration system of claim 12, wherein the adjustment module is configured to determine a first change in the present configuration relative to the first configuration of the first sleeve using the first and second coordinate systems.

12

14. The tool reconfiguration system of claim 13, wherein: the attachment device is slidably and rotatably attached to the respective rod branch using a second sleeve; the end effector includes a third identifier disposed on the second sleeve; and

the adjustment module is configured to determine a third coordinate system on the second sleeve based on a position of the third identifier.

15. The tool reconfiguration system of claim 14, wherein the adjustment module is configured to determine whether the end effector has been adjusted from the first configuration to the second configuration based on the first, second, and third coordinate systems.

16. The tool reconfiguration system of claim 1, wherein: the photogrammetry module includes a second camera configured to capture second images; and the adjustment module is configured to:

receive the second images of the end effector; and determine whether the end effector has been adjusted from the first configuration to the second configuration based on the images and the second images.

17. A tool reconfiguration method comprising: capturing images of an end effector that is adjustable between a plurality of configurations of suction cups; determining whether the end effector has been adjusted from a first configuration of the suction cups to a second configuration of the suction cups based on at least one of the images; and

in response to a determination that the end effector has been adjusted from the first configuration of the suction cups to the second configuration of the suction cups, selectively actuating one or more components of the end effector and adjusting a present configuration of the suction cups of the end effector to the first configuration.

18. The tool reconfiguration method of claim 17, wherein the determining whether the end effector has been adjusted from the first configuration to the second configuration includes:

determining a first coordinate system based on a position of a first identifier that is disposed on the end effector and captured in the at least one of the images;

determining a second coordinate system based on a position of a second identifier that is disposed on the end effector and captured in the at least one of the images; and

determining whether the end effector has been adjusted from the first configuration to the second configuration based on a change in the first and second coordinate systems.

19. The tool reconfiguration method of claim 17, further comprising:

storing the second configuration in memory in response to receipt of user input indicating the end effector was intentionally adjusted from the first configuration to the second configuration.

20. The tool reconfiguration method of claim 18, wherein each of the first and second coordinate systems includes an origin point, an X axis, a Y axis, and a Z axis.